Neuromorphic based classifier for Alzheimer classification with EEG data - project results

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Abstract

This study investigates the feasibility of using Spiking Neural Networks (SNNs) for the classification of Alzheimer's Disease (AD), Mild Cognitive Impairment (MCI), and Healthy Controls (HC) based on spectral EEG features. Motivated by the clinical need for energy-efficient and portable diagnostic tools, we propose a shallow SNN architecture utilizing Leaky Integrate-and-Fire (LIF) neurons trained via surrogate gradient learning. The methodology incorporates a rigorous preprocessing pipeline, including stratified data splitting, ANOVA-based feature selection (Top-K), and Synthetic Minority Over-sampling Technique (SMOTE) to address class imbalance. We evaluated two temporal encoding strategies: Rate Encoding and Latency Encoding. Preliminary results demonstrate that while Rate Encoding achieves superior overall stability (AUC $\approx$ 0.85), Latency Encoding offers higher sensitivity for AD detection (Recall 60%). The study confirms the potential of neuromorphic computing in processing biomedical signals, highlighting the trade-off between classification stability and temporal precision, and identifies the differentiation of MCI as a critical challenge for future research.

Keywords: Spiking Neural Networks (SNN), Alzheimer's Disease, EEG Signal Processing, Neuromorphic Computing, Latency Encoding, Rate Encoding, Class Imbalance, SMOTE, Surrogate Gradient Learning.

1 Introduction and review of related research

1.1 Motivation and Rationale

1.1.1 Limitations of Conventional Computing Approaches

A widely recognized limitation of traditional deep learning architectures is their significant computational demand. This inefficiency stems from several intrinsic factors. The training process typically requires iterative optimization over high-dimensional parameter spaces using backpropagation and gradient-based methods, which is computationally intensive. The memory footprint for storing model parameters and activations can also be prohibitive, limiting deployment on resource-constrained hardware. These factors collectively result in high energy consumption and increased operational, presenting a barrier for practical applications where computational resources are limited [1].

Scaling of deep learning models frequently relies on cloud-based computing infrastructure, which incurs substantial energy costs. The computational intensity of training and serving large models translates directly to high power draw from high-performance computing (HPC) clusters, particularly from GPUs and associated cooling systems. This energy demand contributes to a significant carbon footprint for large-scale AI operations. Furthermore, the energy overhead is compounded by the continuous operation of data centers required for model serving, even during periods of low inference demand, leading to inefficient resource utilization from a power perspective [1].

Additionally, despite their nomenclature, artificial neural networks (ANNs) exhibit a significant divergence from their biological counterparts. Their fundamental mechanisms, such as the backward pass of error signals via backpropagation, have no identified equivalent in neurobiological systems. Furthermore, typical ANN neurons employ simplified activation functions and precise, synchronous update cycles that do not reflect the complex, asynchronous, and dynamic spiking behavior of biological neurons. The common use of global weight updates and dense, layered connectivity also stands in contrast to the localized, sparse, and plastic nature of synaptic learning in the brain. This gap limits the utility of ANNs as models for studying neural computation [2].

1.1.2 Clinical Needs and Practical Requirements

An early and accurate diagnosis of Alzheimer's disease (AD) is a critical clinical need. It is vital for patients to access the right pharmacological and non-pharmacological interventions at the right time, especially new disease-modifying therapies (DMTs) whose efficacy is limited to the early stages (MCI or mild dementia). An early diagnosis also allows patients and their families to make future care plans. Furthermore, economic modeling shows that interventions that delay disease progression can have a significant positive impact on the economic burden of the disease [3].

AD is a progressive disorder, making longitudinal monitoring essential to track disease progression and treatment effects. While gold-standard tools like PET and CSF are invasive and expensive, EEG is a non-invasive, cost-effective alternative suitable for repeated use. Studies show that EEG measures of "oscillatory slowing" (e.g., increased theta power, decreased beta power) are sensitive markers of disease progression, with the fastest rates of change observed in patients at the MCI stage. This makes EEG an ideal tool for monitoring patients within the precise therapeutic window [4].

The primary limitations of "gold standard" diagnostics like advanced neuroimaging (PET) and cerebrospinal fluid (CSF) biomarkers are their high cost, invasiveness, and limited accessibility, which restrict their use for universal or routine screening. This creates an urgent clinical need for non-invasive, cost-effective, and accessible screening tools. Quantitative EEG (qEEG) is a leading candidate to fill this gap, driving development towards portable, affordable systems that can be used in primary care settings [5].

For successful clinical adoption, new AI-based diagnostic tools must be integrated into existing clinical workflows. This presents significant challenges, including the need to navigate regulatory barriers and address ethical concerns about patient data. A primary hurdle is the "black box" nature of many complex AI models. This "lack of interpretability" is a critical limitation, as it erodes clinical trust and limits applicability in high-stakes diagnostics [6], [7].

1.1.3 Potential of Neuromorphic-EEG Combination

Neuromorphic processors offer a native architectural compatibility for processing the temporal dynamics in EEG signals. These hardware platforms are designed around event-driven, asynchronous computation, which aligns with the sparse, time-varying nature of neural activity recorded by EEG. The fundamental processing element in neuromorphic systems is the spiking neuron, which inherently models temporal integration and leaky memory, providing a natural substrate for capturing the multiscale temporal dependencies found in brain signals. This correspondence eliminates the need for the continuous, high-frequency polling of signal values required by conventional von Neumann architectures, thereby avoiding significant energy overhead. The event-based operation of neuromorphic hardware can map directly onto the temporal structure of EEG, enabling efficient, low-power inference [8].

Event-driven computation provides an efficient approach to processing the sparse, non-stationary nature of neural data. Unlike conventional systems that operate on a fixed clock cycle, neuromorphic processors update internal states only upon the arrival of input spikes. This fundamental asynchrony ensures that computational and energy resources are expended proportionally to the information content of the signal [9].

1.2 Current Research Status

1.2.1 Traditional Machine Learning Approaches

Traditional machine learning (ML) pipelines for AD classification rely on neurophysiological feature extraction. These fall into three main families:

- Spectral features, analyzing band power (e.g., increased slow-wave, decreased fast-wave) to quantify the characteristic "oscillatory slowing" [10], [11];
- Complexity features, using metrics like Multi-scale Entropy (MSE) or fractal dimension, which capture the reduced dynamic complexity of AD brain activity [12];
- Functional connectivity features, using metrics like the Phase Lag Index (PLI) to measure the loss of synchronization between brain regions.

The extracted feature vectors are then fed into conventional classifiers. Support Vector Machines (SVM) are a dominant choice, demonstrating high performance, with studies achieving a notable accuracy of 96.01 % on spectral and statistical features. Random Forests (RF) are also widely used and valued for their robustness; studies using RF classifiers to distinguish AD patients from healthy controls have reported excellent performance, including 98% sensitivity and 98% specificity [13].

To bypass manual feature engineering, end-to-end deep learning (DL) is commonly used. Convolutional Neural Networks (CNNs) are the most frequent architecture, applied to raw time-series or 2D spectrogram representations. Hybrid architectures, such as those combining CNNs with Recurrent Neural Networks (RNNs), are also effective at extracting both spatial and temporal features. More recently, Transformer-based models, including hybrid CNN-ViT (Vision Transformer) architectures, are being explored for their ability to capture complex, global patterns in EEG data [14], [15].

While both conventional and DL models report high accuracy, they face critical limitations for clinical deployment. DL models are computationally intensive and have high energy consumption, making them "unsuitable for resource-constrained portable BCI devices". Furthermore, they suffer from the "black box" problem, offering little interpretability into how they make decisions, which is a "critical requirement in medical applications". Finally, these models "often require large amounts of labeled data for training," which are "difficult to obtain in clinical settings" [16].

1.2.2 Neuromorphic Computing in Biomedical Applications

Several neuromorphic hardware platforms have been developed, representing critical milestones in the field's progression. Architectures such as Intel's Loihi, SpiNNaker, and IBM's TrueNorth have demonstrated capabilities for solving computational tasks with low power and latency. These systems often implement event-driven, asynchronous communication protocols, such as address-event representation, to support sparse, spike-based computation. A key feature of platforms like Loihi 2 and SpiNNaker2 is their support for dynamic reconfigurability and neuronal scalability, enabling them to simulate neural behavior across multiple levels of biological detail, from point neurons to complex synaptic models. The scalability of these systems is frequently achieved by interconnecting multiple chips or boards, with some contemporary architectures supporting networks of hundreds of millions of neurons [1].

Neuromorphic computing platforms have demonstrated significant potential for real-time biomedical signal processing and pattern recognition, enabling low-power, embedded diagnostic and therapeutic systems [17]. The core advantage of neuromorphic approaches in these applications is their departure from the von Neumann bottleneck of traditional computing. By co-locating memory and processing and using event-driven, asynchronous computation, these systems achieve high efficiency . This is exemplified by reservoir computing, a neuromorphic approach where a fixed, non-linear system (the "reservoir") projects input signals into a high-dimensional space, making them easier to classify with a simple, trainable output layer . This method is highly effective for pattern recognition in complex, time-varying signals like ECGs and EEGs [18].

1.2.3 Research in Neuromorphic EEG Processing

Emerging research is exploring diverse spike-based encoding methods to convert continuous EEG signals into sparse, event-based spike trains suitable for SNN processing, with applications in detecting pathological states such as epileptic seizures. Rate-based schemes, which encode signal amplitude as firing frequency, provide a straightforward transformation but may sacrifice temporal precision. In contrast, temporal coding schemes like the Bens Spiker Algorithm (BSA) are being applied to map the original EEG data directly into spike timing [8], bypassing conventional feature extraction and selection stages to mitigate associated information loss. This approach aims to preserve the millisecond-scale temporal dynamics inherent in EEG, which are critical for distinguishing seizure activity. The encoded spike trains are subsequently processed by multilayer feedforward spiking neural networks, which can be trained with supervised learning algorithms based on desired spike train reconstruction. For example, Lin et al ([19], 2024) used this approach on benchmark datasets like the CHB-MIT scalp EEG corpus, achieving classification accuracy of over 95%.

Current research into Spiking Neural Network architectures for temporal pattern recognition focuses on designs that leverage their inherent capacity to process spatio-temporal data directly from raw signals, bypassing hand-crafted feature extraction. For example, Behrenbeck et al. ([18] 2019) succesfully used a NeuCube model implemented on SpiNNaker architecture to classify between different grasping patterns directly from raw surface electromyography signals as well as an estimator for motor imagery movements from EEG signals, achieving a test accuracy of 84.8% (of grasping patterns from sEMG signals), a 19% relative mean square error (of estimation of grasping force) and a 75% accuracy (of motor imagery movements from EEG signals).

The application of Spiking Neural Networks (SNNs) to Alzheimer's Disease (AD) analysis is an emerging field that seeks to leverage the energy efficiency and biological plausibility of neuromorphic computing for medical diagnostics. Recent research has demonstrated the potential of SNN-based architectures in classifying AD and its prodromal stage, Mild Cognitive Impairment (MCI), from neuroimaging data.

Turkson et al. ([20] 2021) proposed a foundational deep learning framework for AD classification using MRI scans, which integrated an unsupervised convolutional SNN for feature extraction with a supervised Deep Convolutional Neural Network (DCNN) for final classification. Their methodology employed a time-to-first-spike coding strategy to convert preprocessed MRI data into spike trains, which were then processed by a multi-layer SNN with spike-timing-dependent plasticity (STDP) to learn discriminative feature filters. This hybrid model, implementing a three-binary classification task (AD vs. NC, AD vs. MCI, NC vs. MCI), demonstrated that unsupervised pre-training with SNNs could enhance the discriminative capability of subsequent supervised learning, reportedly outperforming several classical machine learning baselines.

In a related approach utilizing electrophysiological data, a study by Capecci et al. ([21] 2016) employed the NeuCube architecture, an evolving spatio-temporal data machine (eSTDM), to model EEG data from AD and MCI patients. This framework was designed to analyze changes in neural activity across different brain regions, capitalizing on the inherent capability of SNNs to process spatio-temporal information. The model aimed not only to study AD progression but also to predict the conversion from MCI to AD, showcasing the applicability of SNN architectures to dynamic, time-series biomedical signals for prognostic purposes.

More recently, Wu et al. ([22] 2025) introduced FasterSNN, a hybrid architecture designed to address limitations in SNN expressiveness and training stability for 3D MRI classification. This model integrates Leaky Integrate-and-Fire (LIF) neurons with a region-adaptive convolution strategy and a multi-scale spiking attention mechanism. The design emphasizes computational efficiency, achieving competitive classification accuracy on the ADNI and AIBL datasets while significantly reducing energy consumption compared to conventional Artificial Neural Networks (ANNs) and Transformer models. The work highlights a trend towards developing more sophisticated and efficient SNN architectures that are practical for clinical deployment.

One of the primary challenges in training Spiking Neural Networks (SNNs) is that their discrete, all-or-nothing spike events are non-differentiable functions. This makes it difficult to use standard gradient-based backpropagation, which is the engine of modern deep learning [23].

A highly effective and pragmatic methodology to overcome this is ANN-to-SNN transfer learning. This approach provides a bridge, combining the best of both paradigms. The method involves first training a conventional Artificial Neural Network (ANN)—such as a CNN—on the EEG data using standard backpropagation. Once trained, its network weights are transferred or converted to an SNN, which typically has an identical architecture [17].

The primary advantage is that the SNN "inherits the learned features" from the high-performance ANN, bypassing the difficult SNN-specific training problem. This allows the model to achieve the high accuracy of a deep ANN while gaining the "significant energy savings" and "reduced computational complexity" of a neuromorphic SNN. For example, a study on EEG signal classification using this method achieved 90% accuracy while consuming only 0.077 W for inference. This technique is a robust and widely reviewed pathway for developing classifiers for biosignals like EEG [24].

1.2.4 Research Gaps and Opportunities

The analysis of clinical needs and the state of the art reveals several critical research gaps, which represent the primary opportunities for this project.

The central justification for this project stems from the limited exploration of neuromorphic systems for this specific task. While Spiking Neural Networks (SNNs) are recognized for their potential to identify biomarkers for neurological disorders like Alzheimer's from EEG data, the body of published literature specifically targeting AD-EEG classification is sparse. The few existing studies are largely dominated by specific, evolving SNN architectures (e.g., NeuCube), indicating that the field is not yet broadly explored with modern, standardized SNN models [25], [21].

A primary technical hurdle involves the challenge of signal-to-spike conversion. SNNs process discrete spikes, but EEG is a continuous, analog, and notoriously non-stationary signal. The process of converting this signal into a spike train is a critical bottleneck. Current SNN solutions for EEG often suffer from "inefficient spike encoding" , which can lead to information loss and fails to preserve the rich temporal information that is clinically relevant. The opportunity is to investigate encoding strategies (see Table 1) that optimally preserve the discriminative features of the EEG signal [3], [26].

Table 1: Comparison of Common Signal-to-Spike Encoding Techniques

Encoding Method	**Description**	**Potential Challenge for EEG**
Rate Coding	Firing rate is proportional to signal amplitude.	Temporally inefficient; may lose precise timing information critical for EEG.
Temporal Coding	Information is encoded in the *timing* of spikes.	Can be highly sensitive to the low signal-to-noise ratio inherent in EEG.
Delta Modulation	Spikes are generated when the signal changes by a threshold.	Can be sensitive to high-frequency noise, leading to inefficient spiking.

A significant methodological hurdle is the need for standardized evaluation frameworks. The neuromorphic field currently "lacks standardized evaluation benchmarks". This fragmentation "complicates direct model comparisons and hinders systematic scientific progress". This problem is systemic because traditional AI benchmarks like FLOPS (Floating-point Operations Per Second) are "ill-suited" and "ineffective" for measuring the performance of event-driven, asynchronous SNNs. The opportunity is to evaluate SNNs on the metrics that justify their existence: not just accuracy, but also energy-efficiency (e.g., synaptic operations) and latency. New, comprehensive benchmarks are needed to enable fair comparisons [7].

A key implementation hurdle stems from the "black box" nature of many complex AI models. Their "lack of interpretability" is a critical limitation that restricts "clinical applicability". This is a significant barrier, as it "weakens the trust of clinicians" who are hesitant to base high-risk decisions on opaque systems. The opportunity is to leverage the bio-inspired nature of SNNs. By "integrati[ng] domain-specific knowledge" into the model architecture, SNNs may offer a more interpretable "grey box" solution, enhancing clinical trust [27].

The primary clinical validation hurdle is scalability and generalization across patient populations. A significant challenge for all EEG-based AI models is generalization. Many high-performing models are "patient-specific" (subject-dependent). These models are "limited by their reliance on individualized calibration, which restricts their scalability for practical applications". A clinically viable tool, however, must be "subject-independent" (or "cross-patient"), capable of performing robustly on new patients. Demonstrating this cross-patient generalization is "of great significance" for any clinical tool and remains a critical, underexplored challenge [28], [29].

2 Problem formulation and proposed solution

2.1 Algorithm Selection

The core computational model selected for this project is a **Spiking Neural Network (SNN)**. This choice is motivated by the need for an energy-efficient, temporally-aware, and potentially more interpretable classifier, aligning with the clinical and practical requirements identified in Section 1. The model will process precomputed spectral EEG feature vectors per participant and perform multiclass classification into Alzheimer's Disease (AD), Mild Cognitive Impairment (MCI), and

Healthy Control (HC) categories. A shallow, fully-connected architecture based on Leaky Integrate-and-Fire (LIF) neurons is proposed to balance performance with computational simplicity and to mitigate overfitting on limited clinical datasets.

2.1.1 Motivation for SNN choice

Energy Efficiency: SNNs process information only when spikes occur, leading to significantly lower power consumption compared to continuously active artificial neural networks (ANNs). This aligns with the need for portable, battery-operated EEG diagnostic systems.

Temporal Processing: EEG-derived spectral features contain implicit temporal dynamics. SNNs natively handle time-encoded information, making them suitable for capturing progression-related changes in Alzheimer's disease.

Bio-Inspired Interpretability: The spike-based communication resembles neural activity, offering a more transparent framework for linking model decisions to neurophysiological events, addressing the "black-box" limitation of deep learning models.

Hardware Compatibility: SNNs can be efficiently deployed on emerging neuromorphic processors (e.g. SpiNNaker), enabling real-time, low-latency inference in clinical settings.

2.1.2 Proposed architecture

The proposed SNN follows a shallow, feedforward structure:

- *Input Layer:* One spiking neuron (or a small population) per spectral feature.
- *Hidden Layer(s):* One or two layers of LIF neurons with adaptive synaptic weights.
- *Output Layer:* Three neurons representing the classes: AD, MCI, and HC.
- *Connectivity:* Fully connected between layers, with optional sparse connections to promote efficiency and avoid overfitting.

2.1.3 Input representation

The model utilizes precomputed spectral EEG features, comprising band power values (e.g., delta, theta, alpha, beta) per channel and inter-band power ratios, which may be augmented with derived metrics such as asymmetry or coherence. These static feature vectors are z-score normalized across the training set to ensure stable spike encoding. For input to the spiking neural network, the normalized vectors are converted into temporal spike trains via rate or latency encoding and presented over a defined simulated time window.

2.1.4 Output and decision mechanism

The output and decision mechanism for classification is based on the spiking activity within the final layer of the network. Over a fixed simulation period, the spikes generated by each output neuron are accumulated to compute a firing rate for each class-specific unit. The final classification decision follows a simple winner-take-all rule, whereby the class label corresponding to the neuron exhibiting the highest firing rate is selected. To augment this discrete decision with a measure of certainty, the firing rates are normalized, typically via a softmax function, transforming them into a probabilistic confidence distribution across all possible classes. This approach provides both a definitive classification and a continuous confidence metric for each prediction.

2.2 Optimalization problem

The core optimization challenge in this project involves training a Spiking Neural Network (SNN), widely regarded as the third generation of neural network models [30], to classify complex EEG patterns. Unlike standard Artificial Neural Networks (ANNs), SNNs operate on discrete, non-differentiable events (spikes). This introduces the Temporal Credit Assignment Problem, which necessitates determining how past neuronal activities contribute to the current classification error in a dynamic system [31].

2.2.1 Encoding Strategy

The optimization process begins with the transformation of static spectral features into the temporal domain. Since the efficiency of the SNN depends heavily on the input sparsity, we will evaluate two distinct encoding paradigms:

1. *Rate Encoding:* the normalized amplitude of each spectral feature is linearly mapped to the firing frequency of the corresponding input neuron over a fixed time window T. While robust to noise, this method is computationally expensive due to the high number of generated spikes.
2. *Latency Encoding (Time-to-First-Spike):* information is encoded in the precise timing of the first spike. Higher feature values result in earlier firing times. This approach maximizes temporal precision and energy efficiency but requires strict noise control.

The selection of the encoding strategy is treated as a hyperparameter optimization problem, balancing classification accuracy against the total spike count (energy consumption).

2.2.2 Objective Function Formulation

The learning objective is defined as the supervised minimization of a composite Loss Function L_{total} with respect to the synaptic weights $\mathbf{W}$. To align with the requirements for wearable EEG devices [32], the objective function combines classification accuracy with a sparsity constraint.

$$L_{total}(\mathbf{W}) = L_{cls} + \lambda L_{reg} \tag{1}$$

1. *Classification Loss (L_{cls}):* we employ the *Cross-Entropy Loss* applied to the temporal average of output spikes. Let $S_{out,c}(t)$ be the binary spike output of neuron c at time t, and T be the simulation window duration. The predicted probability $\hat{y}_c$ for class c is derived via the Softmax function:

$$\hat{y}_c = Softmax\left(\frac{1}{T}\sum_{t=1}^{T} S_{out,c}(t)\right) \tag{2}$$

   The loss is strictly defined as $L_{cls} = -\sum_c y_c \log(\hat{y}_c)$, where y_c represents the ground truth label.

2. *Sparsity Regularization (L_{reg}):* To mitigate biologically implausible high-frequency firing and reduce computational cost, we introduce a penalty term based on the mean squared error between the actual firing rate and a target low sparsity level ν_{target}:

$$L_{reg} = \frac{1}{2}\sum_l \sum_i \left(\frac{1}{T}\sum_t S_{l,i}(t) - \nu_{target}\right)^2 \tag{3}$$

2.2.3 Addressing Non-Differentiability: Surrogate Gradient Learning

A fundamental hurdle in optimizing SNNs is the non-differentiable nature of the spike generation mechanism. The Leaky Integrate-and-Fire (LIF) neuron utilizes a Heaviside step function $H(u)$ [33]:

$$S(t) = H(u(t) - V_{th}) \tag{4}$$

The derivative of $H(u)$ is a Dirac delta function, which vanishes everywhere except at the threshold, effectively blocking the gradient flow required for standard Backpropagation ("The Dead Neuron Problem").

To resolve this, we implement the *Surrogate Gradient* method proposed by Neftci et al. [34]. While the *forward pass* retains the exact spiking dynamics, the *backward pass* approximates the derivative $H'(u)$ using a continuous, smooth function $\sigma'(u)$, such as the fast sigmoid derivative:

$$\frac{\partial S}{\partial u} \approx \sigma'(u) = \frac{1}{(1 + |\alpha u|)^2} \tag{5}$$

This approximation enables the application of Backpropagation Through Time (BPTT) and standard optimizers (e.g., Adam) to effectively tune the network weights.

2.2.4 Training Methodology

The optimization is performed using the *AdamW* optimizer, which adapts the learning rates for each parameter. To ensure clinical relevance and generalization capability, the training follows a *subject-independent cross-validation* protocol. This prevents the model from learning subject-specific artifacts rather than disease-specific biomarkers.

3 Experimental research results

3.1 Dataset description

The study utilizes a subset of open-source resting-state electroencephalography (EEG) dataset comprising of 269 records as introduced by Meghdadi et al. [35]. Each record contains 3595 features derived from spectral densities and inter-band ratios calculated across all measurement channels. The cohort includes four distinct populations: patients with Alzheimer's Disease (AD; $n = 26$, age 73.5 ± 8.5), individuals with Mild Cognitive Impairment (MCI; $n = 53$, age 71.2 ± 7.8), and two healthy control groups, HC1 ($n = 129$, age 24.4 ± 4.3) and HC ($n = 55$, age 68.8 ± 5.7). To ensure demographic consistency and mitigate age-related physiological bias, the HC1 group was excluded from the classification task. The final classification target consists of the age-matched AD, MCI, and HC groups.

3.2 Methods

3.2.1 Model architecture

The implementation relies on *PyTorch* for defining network modules, parameter optimization, and tensor operations. The spiking components are provided by the *snntorch* library, which supplies the Leaky Integrate-and-Fire neuron models and surrogate gradient functions used during training.

The model architecture consists of a Spiking Neural Network (SNN) utilizing Leaky Integrate-and-Fire (LIF) neurons across a sequence of fully connected layers. The network is structured with an initial linear transformation, followed by a configurable number of hidden layers, and a final output layer. Each hidden layer incorporates a linear projection and a LIF activation function; and a dropout layer with a 0.5 probability for regularization is introduced between final hidden layer and output layer. To enable gradient-based optimization, the non-differentiable spiking mechanism employs a fast sigmoid surrogate gradient. The simulation proceeds over a fixed number of discrete time steps, where the membrane potential of each neuron is updated according to a decay rate β and an activation threshold θ. The forward pass integrates input features over the temporal dimension, maintaining stateful membrane potentials across steps and returning the accumulated spike recordings and membrane potential traces from the output layer.

The schematic of network architecture is shown in Figure 1.

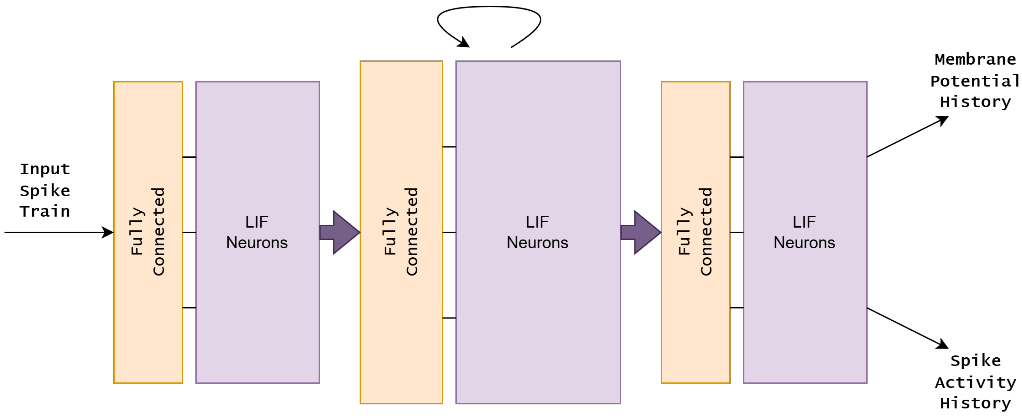

Figure 1: Schematic of SNN architecture

3.2.2 Data processing

The dataset was partitioned into training, validation, and testing sets using a stratified sampling approach to preserve class proportions across subsets. An initial split reserved 20% of the total data for testing, while the remaining 80% was subdivided into training and validation sets using a 75:25 ratio, resulting in an overall distribution of 60% training, 20% validation, and 20% testing. Feature scaling was performed by applying a Min-Max transformation to bound values within the interval [0, 1], with the scaler fitted exclusively on the training partition to prevent data leakage. Dimensionality reduction was executed via univariate feature selection using the ANOVA F-value method to retain the $k = 100$ most significant features. To address class imbalance within the training set, the Synthetic Minority Over-sampling Technique (SMOTE) [36] was applied. Following oversampling, all feature values were clipped to the [0, 1] range to ensure numerical stability of numerical-to-spike encoding methods.

3.2.3 Training pipeline

The training pipeline employs a supervised learning framework optimized via the AdamW algorithm [37]. Input features are converted into the spiking domain using both rate and latency encoding schemes across a fixed temporal window of T steps. For the rate encoding configuration, a weight decay of 10^{-2} is applied, while latency encoding utilizes a reduced decay of 10^{-5}. The optimization objective is defined by a weighted cross-entropy loss function to compensate for class distribution imbalances. During each training epoch, mini-batches of size 32 are processed, where the mean membrane potentials of the output layer serve as the logits for gradient computation. Model performance is evaluated on the validation set at the conclusion of each epoch, with classification accuracy determined by the cumulative spike count of the output neurons. Parameters are iteratively updated through backpropagation through time (BPTT) using surrogate gradients of the network, and the model state corresponding to the highest validation accuracy is preserved for subsequent testing.

3.3 Results

3.3.1 Training history

The training progression for the SNN models utilizing rate-coded and latency-coded data is illustrated in Figure 2. For the rate-coded model, the loss initiates at approximately 1.05 and decreases to a minimum of 0.50 over 50 epochs. The corresponding validation accuracy exhibits an initial increase within the first 10 epochs before fluctuating between 55% and 80% for the remainder of the training duration. In contrast, the latency-coded model begins with a loss value of 1.22 and declines to approximately 0.70 by epoch 50. The validation accuracy for the latency-coded data starts at 28% and demonstrates a stepwise upward trend, plateauing intermittently near 50% between epochs 10 and 30 before reaching a maximum value of 74%. Both models show a general convergence in loss, though the rate-coded model maintains higher peak validation accuracy with greater epoch-to-epoch variance.

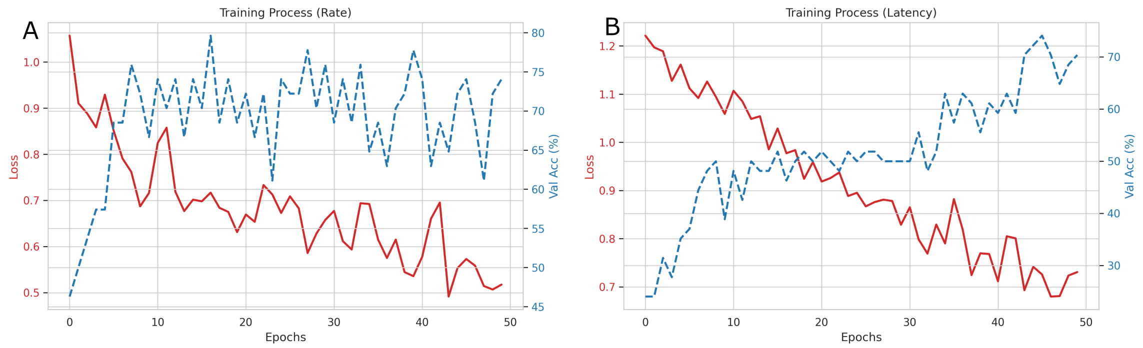

Figure 2: Validation losses and validation accuracies during model training for rate encoded [A] and latency encoded [B] data

3.3.2 ROC Curves

The Receiver Operating Characteristic (ROC) curves for the classification task is presented in Figure 3. The rate-coded model achieves Area Under the Curve (AUC) values of 0.85 for AD, 0.80 for HC, and 0.77 for MCI. The latency-coded model yields AUC values of 0.77 for AD, 0.84 for HC, and 0.76 for MCI. In both encoding schemes, all class-specific curves remain above the diagonal chance line. The rate-coded approach demonstrates the highest individual class performance for AD classification, while the latency-coded approach produces the highest individual class performance for the HC group. The MCI class consistently yields the lowest AUC values across both encoding methodologies.

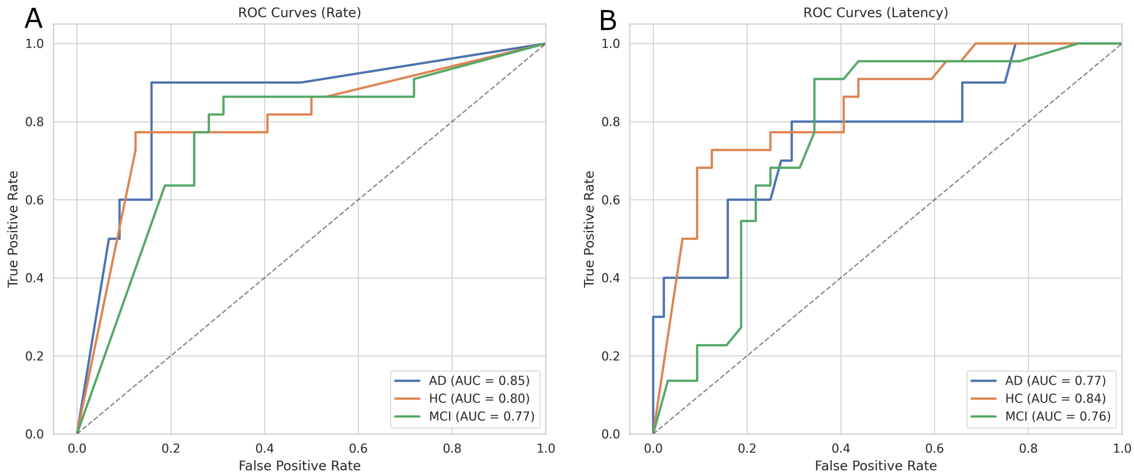

Figure 3: Receiver operating characteristic curves for models using rate encoded [A] and latency encoded [B] data

3.3.3 Performance on test set

The classification performance on the test set for both rate-coded and latency-coded SNN models is summarized through normalized confusion matrices in Figure 4.

For the rate-coded model, the average F1-score is 0.68, average recall is 0.69 and average precision is 0.68. In the latency-coded model, the average F1-score is 0.62, average recall is 0.63 and average precision is 0.61. Classification performance metrics for both encoding schemes are constrained by the limited discriminative accuracy between the intermediate Mild Cognitive Impairment (MCI) group and the remaining classes.

Comparatively, the rate-coded model achieves the highest individual class recall (0.77 for HC) and precision (0.72 for AD). The latency-coded model demonstrates a higher recall for the AD class (0.60) than the rate-coded variant but lower performance in the MCI classification across all metrics. Both models show the highest classification consistency for the HC group.

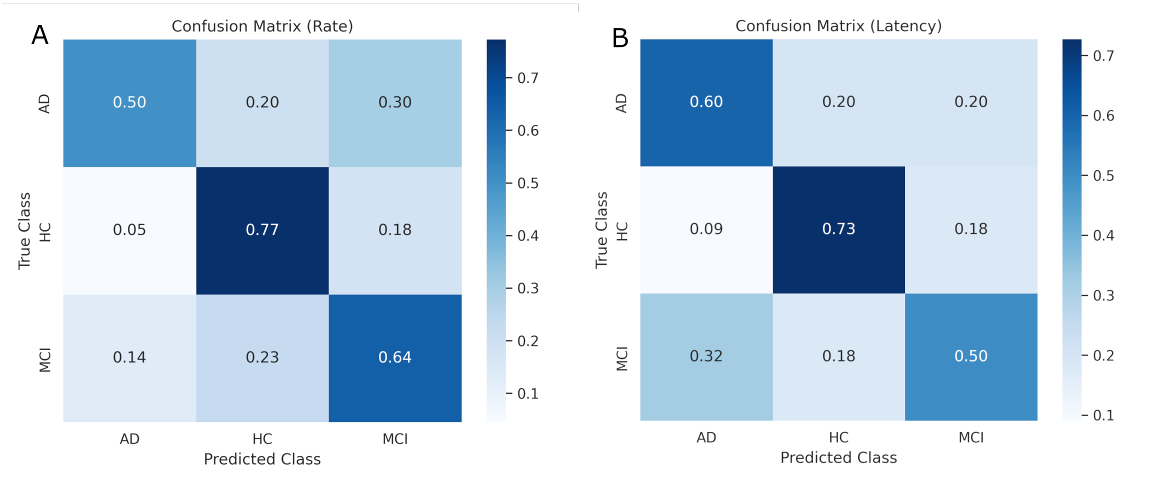

Figure 4: Confusion matrices of predictions on testing set for models using rate encoded [A] and latency encoded [B] data

4 Summary and Conclusions

The primary objective of this thesis was to develop and evaluate a neuromorphic-based classifier capable of detecting Alzheimer's Disease (AD) and Mild Cognitive Impairment (MCI) using spectral EEG features. By leveraging Spiking Neural Networks (SNNs), the study aimed to address the growing clinical demand for energy-efficient, interpretable, and portable diagnostic tools. The research successfully validated the feasibility of this approach, offering critical insights into the trade-offs between different temporal encoding strategies.

4.1 Feasibility of Shallow Neuromorphic Architectures

We demonstrated that deep architectures are not strictly necessary for effective EEG classification in this domain. A shallow SNN architecture, comprising a single hidden layer with approximately 100 Leaky Integrate-and-Fire (LIF) neurons, proved sufficient to extract discriminative biomarkers from high-dimensional spectral data. The model utilized learnable membrane parameters (β decay and threshold), allowing the network to adapt its physical dynamics to the signal characteristics during the training process via surrogate gradient learning. This minimalist design directly contributes to computational efficiency, making it a viable candidate for deployment on resource-constrained edge devices.

4.2 Performance Trade-offs: Rate vs. Latency Encoding

A comparative analysis of the two encoding paradigms revealed distinct performance characteristics, highlighting a trade-off between stability and sensitivity:

- **Rate Encoding (Stability):** This method achieved superior overall discrimination capability, yielding an Area Under the Curve (AUC) of **0.85** for the AD class. It proved to be more robust against noise, providing stable training convergence.

- **Latency Encoding (Sensitivity):** While achieving a lower AUC for the AD class (**0.77**), this time-to-first-spike approach demonstrated significantly higher sensitivity (Recall) for detecting Alzheimer's Disease (**60%** vs. **50%** for Rate Encoding). In the context of medical screening, where minimizing false negatives is crucial, Latency Encoding offers a strategic advantage despite its lower overall specificity.

However, both methods encountered difficulties in distinguishing Mild Cognitive Impairment (MCI) from Healthy Controls (HC) (AUC $\approx$ 0.76–0.77 for MCI), reflecting the inherent spectral overlap and the subtle nature of biomarkers at the prodromal stages of the disease.

4.3 Methodological Integrity

To ensure the reliability of the reported results, this study implemented a rigorous data processing pipeline designed to eliminate data leakage. Stratified data splitting was performed prior to any feature scaling or selection. Furthermore, the Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively to the training subset. This protocol guarantees that the test results reflect the model's true generalization capability on unseen, imbalanced clinical data, rather than artifacts of improper preprocessing.

4.4 Future Directions

Based on the identified limitations and potential, future work will focus on the following key areas:

1. **Hybrid Encoding Schemes:** Developing a hybrid input layer that combines the noise robustness of Rate coding with the temporal precision and sparsity of Latency coding to optimize both AUC and Sensitivity simultaneously.
2. **Feature Expansion:** The current spectral features may be insufficient for robust MCI detection. Integrating connectivity metrics, such as coherence or phase synchronization, could provide the additional information needed to separate MCI from healthy aging.
3. **Hardware Implementation:** The ultimate validation of energy efficiency requires porting the trained model to physical neuromorphic hardware (e.g., SpiNNaker or Intel Loihi). Real-world measurements of power consumption (watts) and inference latency will be essential to confirm the practical utility of the proposed solution in wearable diagnostics.

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